



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)
BSc CHEMISTRY**

Programme	B.Sc Chemistry				
Course Title	INORGANIC CHEMISTRY-VI				
Type of Course	MAJOR/MINOR				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-		60
Pre-requisites	Bonding in Coordination compounds Classification of ligands Bonding in CO molecule Basic idea of IR spectroscopy Metal ions in biological systems				
Course Summary	This course explains in detail the structure, bonding and reactions of organometallic compounds. It deals with the bonding in metal carbonyls and provides application skill in evaluating the bonding and structural characteristics of metal carbonyls using IR spectroscopy. It identifies the application of organometallic compounds. It describes different organometallic polymers. It evaluates bioinorganic compounds and their biological actions				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Equip with comprehensive understanding of organometallic compounds	U	C	Instructor-created exams / Assignments/Quiz
CO2	Identify bonding, synthesis, reactions and applications of metal carbonyls and apply IR spectroscopy to analyse structure and bonding characteristics of metal carbonyls	An	P	Assignment / seminar/quizzes/Class test

CO3	Apply organometallic compounds in synthetic chemistry	Ap	C	Assignment/Seminar/Class test
CO4	Provide with deep understanding of the interplay between bio inorganic compounds and biological systems	U	C	Class Test/ Assignment/Viva Voce
CO5	Identify and distinguish different categories of organometallic polymers and understand their applications	U	C	Assignment/class test/Seminar
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Mark
I	ORGANOMETALLIC COMPOUNDS		10	21
	1	Organometallic compounds. Classification and nomenclature.	1	
	2	Zeise's salt, The 16 and 18 electron rules. Electron counting-covalent and ionic models	1	
	3	Main group organometallics-alkyl and aryl groups 1, 2, 12, 13, 14 and 15, Synthesis, Structure and Applications.	2	
	4	Transition metal to carbon multiple bond-the metal carbenes and carbynes.	2	
	5	Transition metal complexes with chain π ligands-synthesis, structure, bonding and reactions of complexes of ethylene, allyl, butadiene and acetylene.	4	
II	METAL CARBONYLS		15	31

	6	Metal carbonyls- Bonding modes of CO.	1	
	7	IR spectroscopy as a tool to study bonding and structure of metal carbonyls.	1	
	8	Synthesis of Metal carbonyls, Direct and reductive Carbonylation	3	
	9	Reactions of Metal carbonyls-Activation of metal carbonyls,	2	
	10	Disproportion, Nucleophilic addition, electrophilic addition to the carbonyl oxygen, Carbonyl cation, anions and hydrides	3	
	11	Collmann's reagent, Migratory insertion of carbonyls	2	
	12	Oxidative decarbonylation. Photochemical substitution. Microwave assisted substitution.	3	
III	APPLICATIONS OF ORGANOMETALLIC COMPOUNDS IN ORGANIC SYNTHESIS AND HOMOGENEOUS CATALYSIS		11	22
	13	Complex formation and activation of H ₂ , N ₂ , O ₂ , NO by transition metals	3	
	14	Catalytic steps, Oxidative addition, Reductive elimination and Insertion reactions.	2	
	15	Hydrozirconation of alkenes and alkynes	1	
	16	Homogeneous catalysis. Hydrogenation, Isomerization of alkenes, alkyne, Cycloadditions, Ziegler-Natta catalysis	3	
	17	Hydroformylation of alkenes, Monsanto acetic acid process and Wacker process. Metal complexes in enantioselective synthesis	2	
IV	BIOINORGANIC COMPOUNDS AND THEIR FUNCTIONS		12	24
	18	Metallo enzymes, Iron enzymes: Structure and functions of Cytochrome P-450, catalase and peroxidase	3	
	19	Copper enzymes: Oxidase, superoxide dismutase and tyrosinase.	2	
	20	Lewis acid role of Zn (II), Structure and functions of Carboxypeptidase and Carbonic anhydrase	2	
	21	Chlorophyll, Photosynthesis, Photosystem I and II. Nitrogen fixation - Nitrogenases.	3	
	22	Storage and transport of metal ions- ferritin, transferrin and siderophores. Toxic effect of metals	2	
V	ORGANOMETALLIC POLYMERS (Open ended)		12	

		The following topics related with organometallic polymers can be selected by the teacher 1. Polymers with organometallic moieties as pendant groups. Polymers with organometallic moieties in the main chain 2. Condensation polymers based on ferrocene, rigid rod polyynes, Poly (ferrocenyl silane)s and their application 3. Polygermanes and Polystannanes 4. Polymers prepared by ring opening polymerisation 5. Organometallic dendrimers		
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REFERENCES

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3. F.A. Cotton and G. Wilkinson, Advanced Inorganic Chemistry. 5th edition, John and Wiley, 1999.
4. R.S. Drago. Physical Methods in Inorganic Chemistry, 2nd edition, affiliated east west press, 1993.
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6. S. F. A. Kettle, Concise coordination chemistry, Nelson, 1969.
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11. B. Douglas, D. McDaniel, J. Alexander, Concepts and Models of Inorganic Chemistry, Wiley Student Edition, 2006.

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Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2				2	1	2				1		1
CO 2							2				1		2
CO 3				1	1	2					1	1	2
CO 4	2				2	1	3				2	1	1
CO 5	2					1	2				2	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Practical exam (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment/ Seminar/ vivavoce	Practical skill evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		